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Investigating capabilities of intermediaries in short food supply chains: a resource-based view

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Abstract

Short food supply chains (SFSCs) have been presented as a sustainable alternative to unsustainable conventional food supply chains (CFSCs). Here, intermediate SFSCs are most capable of expanding their scale and success. However, it remains unclear which intermediaries can do so sustainably, while upholding SFSCs values. Understanding the types of intermediaries and their capabilities is key to ensuring long-term success in the market. Through a systematic literature review, this paper investigates the type of intermediaries operating in SFSCs and their capabilities. Appropriate exclusion criteria eliminated articles that focused on shortening CFSCs without specifically mentioning SFSCs, as well as articles focusing on direct-to-consumer SFSCs, which did not mention intermediaries. A wide range of synonyms for intermediaries and capabilities was included to ensure a thorough search string. Based on these selection criteria, 65 articles have been selected and analyzed. The capabilities of seven different intermediaries are presented in context of the resource-based view. The capabilities are categorized in operational capabilities and relational capabilities. The outcome allows for the assessment of internal resources and external relationships that could strengthen the intermediaries' operations, ultimately leading to sustainable competitive advantage. The practical implications of this research indicate that certain intermediaries are more suitable than others in obtaining competitive advantage toward CFSC and increasing the volumes moved through SFSCs and allowing for sustainable food supply chains. This is the first review to provide an overview of the most discussed intermediaries in SFSCs and their capabilities creating competitive advantage, allowing for future discussions around sustainable SFSCs.

Keywords: Intermediaries, Short food supply chain, Capabilities, Competitive advantage

Introduction

In the current landscape, food distribution is dominated by large actors involved in long and global food supply chains, also known as conventional food supply chains (CFSCs). These supply chains (SCs) are complex systems that involve the movement of food products from producers to consumers through a series of stages. This system is characterized by its reliance on often global and large-scale operations, involving multiple

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intermediaries. Because of the high number of intermediaries as well as large distances between stakeholders, CFSCs are complex in production, purchasing, transportation, warehousing, handling, packaging, processing, distribution and information processing (Olivares Tenorio et al. 2021; Cui et al. 2024).

While CFSCs often prioritize economic efficiency and productivity and are thus capable of delivering large volumes of food to meet the demand of a growing population, these SCs are often criticized for its unsustainability (Gori and Castellini 2023; Mancini et al. 2019; Zoll et al. 2021). Additionally, recent SC disruptions caused by the COVID-19 pandemic and current geopolitical unrest in the world, highlight the vulnerability of CFSCs, especially with regard to global trade (OECD 2020; Caprile 2022; Uliano et al. 2023). These events spark discussion around the development of sustainable food SCs where there is a focus on economic, environmental and social sustainability, while at the same time being able to withstand long-term and short-term external disruptions.

To achieve this, focus is increasingly directed to the development of alternative food SCs, better known as alternative food networks (AFNs) (Renting et al. 2003; Ilbery et al. 2004; Dani 2015). In the past 20 years, researchers have investigated short food supply chains (SFSCs) as part of a solution to create food security and as an alternative to the unsustainable CFSCs. SFSCs are localized food SCs in which stakeholders use direct or intermediate distribution and marketing schemes. These SFSCs differ from CFSCs by operating within the geographical and social framework of a specific area or community, focusing on more localized, sustainable and transparent methods of food production, distribution and consumption (Charatsari et al. 2024; Mustapa and Kallas 2025). Stakeholders in SFSCs aim to create more sustainable, transparent and equitable systems by focusing on environmental responsibility, social fairness and reconnecting producers and consumers (Renting et al. 2003; Michel-Villarreal et al. 2019). SFSCs are social constructions in which stakeholders focus on the development of relationships, trust and social capital between stakeholders and end-consumers rather than solely the exchange of a product (Cleveland et al. 2014; Maciejczak 2014; Corsi et al. 2018; Charatsari et al. 2024).

Sustainability in SFSCs can be achieved in a number of ways. With the exclusion of unnecessary intermediaries, participation in SFSCs allows producers to keep a larger share of the profits and thus get more value for their products increasing economic sustainability (Benedek et al. 2018; Rosol and Barbosa 2021). Social sustainability is increased through socially embedded relationships between producer and consumer which involves the construction of knowledge, value and meaning about the product and its origin, production and consumption (Maciejczak 2014; Malak-Rawlikowska et al. 2019). The existence of SFSCs could furthermore connect rural to urban areas, increase employment, revitalize communities by increasing social cohesion as well as strengthen local cultures and identities (Galli and Brunori 2013). SFSCs support local communities in the face of disruptions and uncertainties and support their goal to obtain food security and resilience by providing a more stable and predictable SC (Foti and Timpanaro 2021; Jia et al. 2024). Finally, while not all producers in SFSCs are focused solely on organic production (Sellitto et al. 2018; Schoolman et al. 2021), local environmental and health sustainability is often improved by shorter food miles and environmentally friendly production methods (Malak-Rawlikowska et al. 2019). Efficient utilization of

SCs would reduce fossil fuels and greenhouse gases (Ferguson Aikins and Ramanathan 2020; Nordmark et al. 2012).

While a possible sustainable solution to CFSC, SFSCs do not address all sustainability issues that occur in the current food systems. For example, production methods and CO_2 emissions are not necessarily more sustainable in SFSCs (Brunori et al. 2016). Furthermore, while SFSCs are often celebrated for reconnecting producers and consumers, they also raise concerns about accessibility and inclusivity. Several studies point out that local food is frequently priced higher than conventional alternatives, which can limit access for lower-income groups and reinforce existing inequalities in food systems (Allen et al. 2003; Allen 2010). This dynamic has been described as part of the "local food trap" (Born & Purcell 2006), where local food is assumed to be inherently more just or sustainable, even though outcomes often depend on the social relations and power structures at play rather than scale itself (Brunori et al. 2016; Landon and Rosol 2025). While some SFSCs may foster social embeddedness and community (Sage 2003; Brunori et al. 2016), they also risk privileging middle-class consumers who have the financial means and cultural capital to participate (Hinrichs 2003). In this sense, SFSCs may unintentionally reproduce exclusionary practices, overlooking the food needs of marginalized populations (Sage 2003).

If SFSCs are to provide a sustainable alternative capable of serving a growing population and withstanding disruptions, they must expand in number and volume. However, despite rising interest, their sales and volumes remain limited, restricting their ability to compete with CFSCs and achieve sustainable growth (Malak-Rawlikowska et al. 2019).

A review of the literature discussing intermediate SFSC shows knowledge regarding the place of intermediate SFSC in food systems, the relationships within the SC governance and the role of intermediate SFSC in obtaining sustainability (Renkema and Hilletofth 2022). However, earlier studies have not yet determined the exact types of intermediaries and their capabilities contributing to competitive advantage in SFSCs. Moreover, it appears to be challenging to sort and classify the many emerging cooperation models functioning as intermediaries in SFSCs, because of the diversity of stakeholders, strategies and organizational forms that do not follow CFSC logic (Maye & Ilbery 2006; Martens et al. 2022). Moreover, many intermediaries operate as informal or socially driven enterprises, complicating systematic analysis (Kump & Fikar 2021; Martens et al. 2022). A categorization of types and capabilities would be necessary to identify which intermediaries in SFSCs are most effective in scaling volumes and ensuring competitiveness against CFSCs. There is some knowledge with regard to the intermediaries that have shown greater capacity to move larger volumes in SFSCs (Malak-Rawlikowska et al. 2019), but little is known about the specific types, resources and capabilities that enable this success. Addressing this gap is essential to understand how intermediaries can drive the long-term growth and sustainability of SFSCs. Therefore, this paper aims to investigate the type of intermediaries operating in SFSCs and their capabilities contributing to sustainable competitive advantage.

To support this aim, the following research questions have been developed: What are the different types of intermediaries operating in SFSCs? And what are their capabilities contributing to sustainable competitive advantage?

The outcome will allow for the assessment of internal resources that could strengthen the intermediaries' operations, ultimately leading to sustainable competitive advantage. This paper contributes to the existing literature on AFN, by providing knowledge regarding the different intermediaries operating in SFSCs and their capabilities. The paper also extends the existing literature by examining these capabilities in the context of the resource-based view (RBV).

The remaining paper is structured as follows: Chapter 2 presents a review of the literature with regard to SFSCs and RBV theory. Chapter 3 presents the systematic literature review (SLR) methodology based on Durach et al. (2017), presented in a PRISMA Flow model. Chapters 4 and 5 present the results of the SLR through a presentation of the research context (4) and the research findings (5). These findings are discussed in Chapter 6 where they are used to answer the research questions. Finally, the conclusions and implications are presented in Chapter 7.

Research background

Resource-based view

The RBV is a strategic management theory that focuses on the internal resources and capabilities of a firm as the primary sources of competitive advantage and superior performance. The firm's resources include tangible resources such as assets, but also intangible resources including capabilities, organizational processes, firm attributes, information and knowledge (Barney 1991). According to this view, firms can achieve sustained competitive advantage when these resources and capabilities are considered valuable, rare, inimitable and non-substitutable (VRIN) (Barney 1991). RBV argues that the combination of these resources and capabilities enables a firm to outperform competitors over the long term, as such advantages are difficult for rivals to replicate, ensuring the firm's continued success and profitability (Christopher 2023). Strategic implications lie in the identification and nurturing of resources and capabilities that meet these VRIN criteria and use them to obtain competitive advantage. The RBV emphasizes the value of internal capabilities more than external market conditions, shifting the focus from competitive positioning in the industry to leveraging unique organizational strengths. While mostly the RBV includes a firm-centric view on managing resources, Christopher (2023) discusses the importance of SC collaboration to obtain necessary resources and capabilities. In a SC context, relationships and networks with suppliers, partners and customers can be considered strategic resources when they are VRIN. The ability to collaborate, coordinate and communicate effectively across firm boundaries is a key capability that supports better SC performance. This includes the alignment of all stakeholders in the SCs with regard to value creation and delivery, and their motivation to reduce overall costs (Christopher 2023). Relational capabilities, such as strategic decision-making, building collaborative networks and creating alliances, enhance the integration and agility of the SC, which are difficult for competitors to replicate (Kim et al. 2015). This focus on SC collaboration is furthermore emphasized in the context of sustainability, where inter-firm resources and capabilities derived from the SC are so ambiguous to understand and socially complex that they are difficult to replicate by competitors, thus ensuring sustained competitive advantage (Gold et al. 2010).

Short food supply chains and competitive advantage

In the literature, SFSCs are discussed often in three types: direct-to-consumer SFSC, in which consumers buy a product directly from the producer on a face-to-face basis, allowing for authenticity and trust via personal interaction; proximate SFSC, where products are sold close to where they are produced, where consumers are aware of the "local" nature of goods at the retail level; and spatially extended SFSC, where the physical distance between producer and consumer is larger, but information about the place of production and the producer is transferred to consumers. The first form of SFSC involves no intermediaries, whereas proximate and spatially extended SFSCs usually involve an intermediary organization (i.e., intermediate SFSC) (Marsden et al. 2000).

While the concept of SFSCs has existed for many years, scholarly and industry interest has intensified considerably since 2014 (Renkema and Hilletofth 2022; Mustapa and Kallas 2025). While direct-to-consumer SFSC previously has been regarded as the most sustainable solution, the disadvantages of these types of SCs are many. In these constructions, producers are not only responsible for producing and processing their goods but are forced to take up additional operational services related to logistics and distribution as well as sales and marketing (Cembalo et al. 2015; Renkema and Hilletofth 2022). Services that otherwise are provided by intermediaries in the SC. These extra activities can be burdensome for small producers and affect their businesses negatively. Additionally, their limited supply and difficulties in scaling up production make them unsuitable suppliers for many business customers, such as hotels, restaurants and catering businesses (HoReCa) as well as public procurement (Lehtinen 2012; Cembalo et al. 2015). Re-introducing intermediaries in SFSCs, creating intermediate SFSCs, can be one way of overcoming the limitations of direct-to-consumer SFSCs, while at the same time offering a way to deal with the rigid and nontransparent conditions of a CFSC (Tewari et al. 2018; Renkema and Hilletofth 2022). Intermediaries SFSCs can provide a wide range of services to both producers and customers in SFSCs that will meet the current growing demand for local food, while simultaneously managing diversified market requirements (Ilbery et al. 2004; Tewari et al. 2018).

Intermediaries in CFSCs are generally motivated solely by the business opportunity of intermediation, without necessarily having intrinsic values focused on relationships with producers and customers connected to participation in SFSCs (Peterson et al. 2021; Tewari et al. 2018). Intermediaries involved in SFSCs often mix both business objectives and mission-driven objectives in their operations, with a focus on local food, community development, health and education. Relationships in the SC are based on close collaborative relationships and trust, where producers are seen as strategic partners rather than substitutable price-driven suppliers (Christopher 2023; Mejia and Garcia-Diaz 2018; Peterson et al. 2021; Tewari et al. 2018).

In SFSCs, competitive advantage depends not only on operational capabilities like management and marketing, but also on relational capabilities such as producer support and community engagement, which are particularly valuable for small and medium producers in rural areas facing shifting market conditions. These relational resources are necessary tools with the emergence of new customers, when existing customers seek additional suppliers, or when product requirements change due to shifting consumer demands (Ilbery et al. 2004; Peterson et al. 2021; Tewari et al. 2018). For these producers

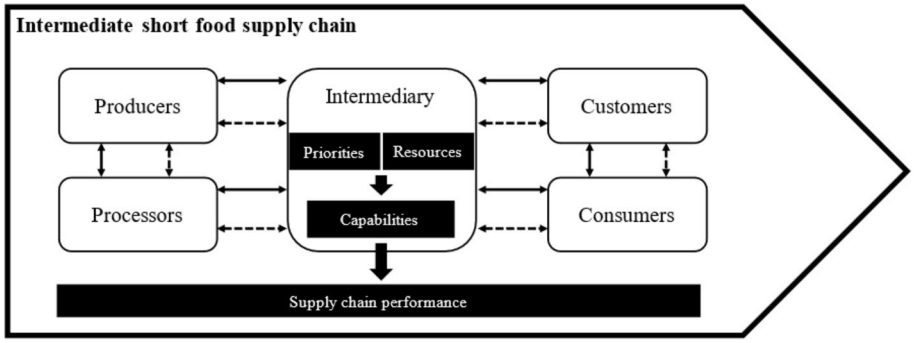


Fig. 1 Capabilities of intermediaries increasing supply chain performance. A resource-based view perspective (own elaboration)

Table 1 Six-step SLR methodology presented by Durach et al. (2017)

Steps	Formulation of steps
Step 1	Formulate the research questions
Step 2	Determine the inclusion/exclusion criteria
Step 3	Retrieve the baseline sample of potentially relevant articles
Step 4	Apply the inclusion/exclusion criteria from step 2
Step 5	Synthesize the articles
Step 6	Report on the results of the SLR

to obtain competitive advantage, they need to connect with key intermediaries. Both the operational and relational capabilities of these intermediaries to create and deliver value in the SC are of importance for the overall sustainability of the SFSCs (Fig. 1). By leveraging both internal resources and external relationships, intermediaries in SFSCs can play a key role in coordinating market transactions (Mejia and Garcia-Diaz 2018).

Methods

SLRs are considered suitable study methods that can provide syntheses of the state of knowledge in a field, if done correctly and transparently (Page et al. 2021). This review has been inspired by earlier published SLRs related to SFSCs, such as, but not limited to, Paciarotti and Torregiani (2021) as well as Renkema and Hilletofth (2022). Correctness was ensured by setting up and following clear guidelines based on the six-step SLR methodology presented by Durach et al. (2017) (Table 1). Transparency of the study was ensured by meticulous reporting using the PRISMA 2020 model (Page et al. 2021). To ensure this study's validity, the authors worked together in close collaboration, while administrating a clear chain of evidence throughout the entire selection and synthesis process. The articles were selected based on predefined criteria (see inclusion/exclusion criteria), and when in doubt or disagreement cross-validated by comparing decisions involving both authors.

The authors retrieved articles from the online databases Web of Science (WoS) and Scopus, with no time restriction, but until the 2nd of December 2024. These are two of the largest and most widely used databases in the social sciences and provide a large variety of peer-reviewed scientific literature. While there exist more databases such as

Google Scholar, this forum is deemed less suitable for research assessment exercises, such as SLRs (Singh et al. 2021). While a large number of scientific articles were included in this review, these databases do not present gray literature or region-specific case studies published in non-indexed outlets.

Step 1 formulate the research questions

Intermediaries in food SCs are important for a wide range of SCM operations. It is, however, unclear which types of intermediaries operate in SFSCs and the capabilities they have that can contribute to sustainable competitive advantage against CFSCs. Therefore, the RQs are formulated as:

- What are the different types of intermediaries operating in SFSCs?
- What are their capabilities contributing to sustainable competitive advantage?

While intermediaries are often conceptualized in diverse ways, this study defines them as organizational entities, rather than mere artifacts, that actively perform intermediary roles in creating value within SFSCs. This perspective aligns with an emphasis on the actors who undertake intermediary functions, rather than on the artifacts or infrastructures through which such functions are enacted.

Step 2 Determine the inclusion/exclusion criteria

This review focuses on food SCs that are not conventional, global or long. There is literature that includes sustainability in these types of SCs, including the focus on shortening the global SC, but they are beyond the scope of this review. According to Corsi et al. (2018), research on AFN includes a wide range of definitions and approaches. This specific review focuses on SFSC research, where the authors selected literature based on the definition by Marsden et al. (2000) specified in the introduction, and other works such as Renkema and Hilletofth (2022). Synonyms for SFSCs can often be found in regional or local food SCs involving a limited number of stakeholders. Therefore, to make the search as broad and relevant as possible, inclusion criteria would include any keyword related to SFSCs such as "local food supply chains" or "alternative food supply chains" and "local food networks" or "alternative food networks." As well as keywords related to intermediaries, such as mediator, aggregator, actor, distributor and their capabilities, such as roles, functions and activities.

Out of scope for this specific study and therefore excluded from the sample are literature related to direct-to-consumer SFSC, other AFN research areas such as urban agriculture including urban farming, or any other form of nonconventional food SC research that do not involve or discuss intermediaries.

A detailed description of the search string can be found in step 3. A full overview of the inclusion and exclusion criteria is shown in Fig. 2.

Step 3 Retrieve the baseline sample of potentially relevant articles

To make the search as thorough as possible, the authors designed a search string that would include terminology covering "short food supply chain," "local food supply chain," "local food network," "alternative food network," alternative food system" and so on by designing the first half of the search string as: ("food supply chain*" OR "food network*" OR "food system*") AND ("local" OR "short" OR "alternative"). The second half of the search string had to focus on the research question that focused on capabilities. The

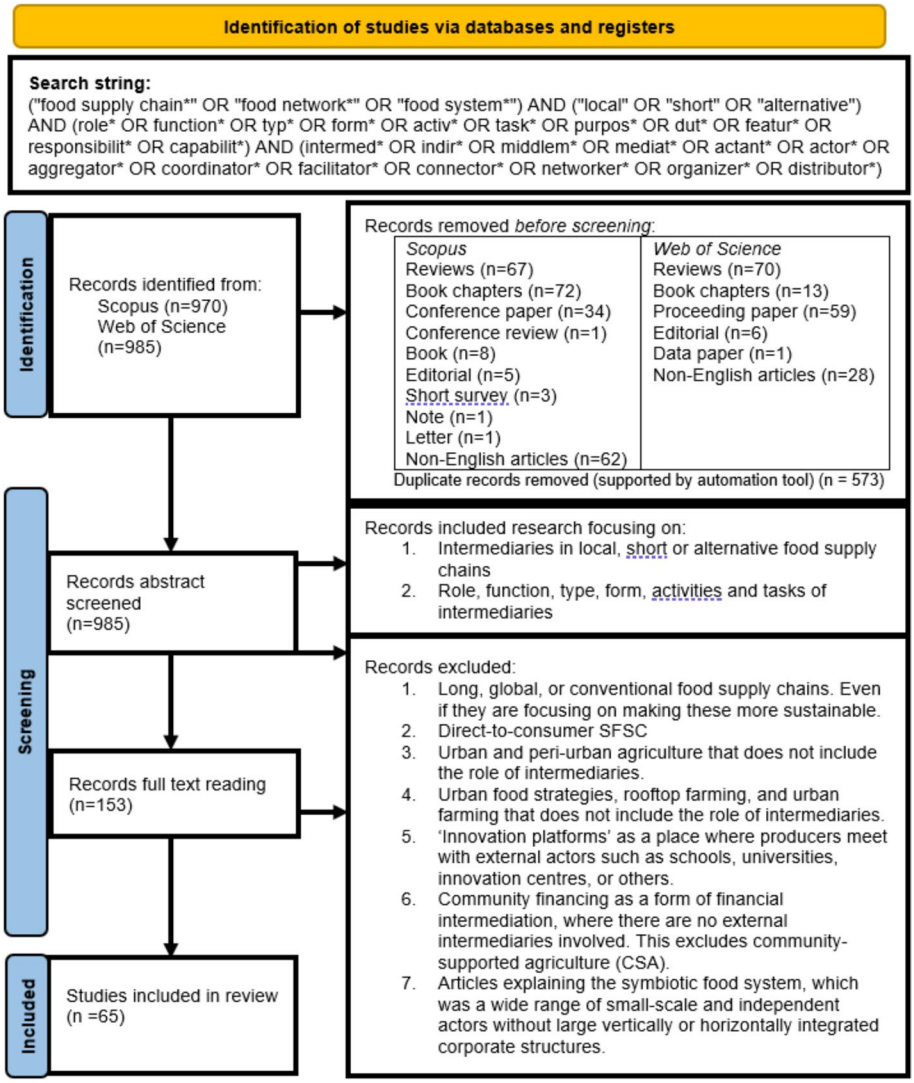


Fig. 2 Systematic Review Process PRISMA Flow Model

capabilities in RBV are appearing in synonyms in different articles. Therefore, to make the search broad enough, the authors included keywords such as "capability," "role," "type," "function," "activity," and "task" and combined them with keywords that reflected on intermediaries by combining several synonyms.

The final search string was constructed using the Boolean operator AND/OR, resulting in:

("food supply chain*" OR "food network*" OR "food system*") AND ("local" OR "short" OR "alternative") AND (role* OR function* OR typ* OR form* OR activ* OR task* OR purpos* OR dut* OR featur* OR responsibilit* OR capabilit*) AND (intermed* OR indir* OR middlem* OR mediat* OR actant* OR actor* OR aggregator* OR coordinator* OR facilitator* OR connector* OR networker* OR organizer* OR distributor*).

The search string was used to search within the "article title, abstract, keywords" in Scopus, and "All Fields" in WoS. The results were filtered on peer-reviewed articles

that were published in journals and only in English, leading to 724 document results in Scopus and 834 in WoS. The results were merged, and duplicates were removed ($n = 573$) using Rayyan software (Ouzzani et al., 2016), leading to a total sample of 985 articles. For full details on the exclusion criteria before screening, see Fig. 2.

Step 4 Apply the inclusion/exclusion criteria from step 2

The authors read the abstracts of 985 articles while focusing on the inclusion and exclusion criteria from step 2, excluding 832 articles. The remaining 153 articles were read in full text while looking for the keywords defined in Step 2 that could help in answering the RQs. The final number of articles included in this review was 65. The authors summarized the systematic review process by following the PRISMA 2020 flow diagram (Fig. 2) (Page et al. 2021).

Step 5 Synthesize the articles.

A deductive approach has been used where the study was designed and analyzed through the RBV lens to find an answer to the RQs. To synthesize the selected articles, the authors used both descriptive and thematic analyses. The descriptive analysis provided an understanding of the research context, while the thematic analysis answered the specific research questions.

The descriptive analysis looked at quantitative elements in the articles, such as the year of publication, country of research conducted and the use of methodologies. The outcome has provided the authors with an understanding of developing trends in research, their geographical focus and statements about the deductive or inductive nature of research (Tranfield et al. 2003).

SFSCs are mostly described as social networks where there are a large group of stakeholders working in different constellations organizing the SC and fulfilling SC capabilities. The authors grouped the different intermediaries based on type and by using the RBV lens the authors looked for capabilities and analyzed similarities and differences between these intermediary types.

During the analysis the authors used computer software to categorize the key intermediaries, and their capabilities found in the selected articles. The authors used Gioia inspired coding structures (Fig. 3) to structure and analyze the data (Gioia et al. 2013). In the lower levels of the coding structure, the researchers have taken an inductive approach to analyze the data. While the overview of coding structure can be derived

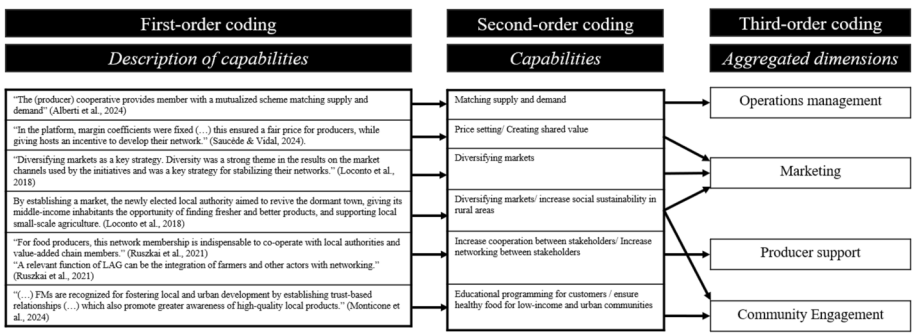


Fig. 3 Example of data coding and aggregation

from the RBV lens, taking a more deductive approach. The final step included the aggregation of capabilities.

Terminology is fragmented in SFSC literature, and both types of intermediaries and the capabilities were refined and improved iteratively. For example, in the case of consumer cooperatives, the literature could be talking about "consumer food networks" or "grassroot initiatives" or "consumer-driven food cooperations." Upon closer study, the authors found their characteristics and capabilities to be the same, leading to the uniform decision to categorize them in one type. The same approach was used with the aggregation of capabilities. Due to the authors' background in SCM and marketing communications, and experience in the use of RBV, the aggregation of capabilities follows a theoretical logic anchored in these subjects.

Step 6 Report the results of the SLR.

Quantitative elements were reported as part of the research context. Qualitative elements are reported in the results. The results were reported using categories developed by the authors based on commonalities in different articles. The thematic analysis looked at qualitative elements in the articles using a thematic coding structure looking for keywords related to types, roles and capabilities, combined with intermediaries (Tranfield et al. 2003).

Research context

The searches in Scopus and WoS had no limitations based on the publication date. All relevant articles, irrespective of the publication time, were included. Among the included sample articles, the oldest article was published in 2004, and the latest articles in the sample were published in 2024 ($n=6$). Most of the articles were published in 2021 ($n=12$), followed by 2022 ($n=9$). While not a linear growth, on average, yearly publications increased from 2014 onwards, illustrating an increased interest in the field.

The research methods of the articles were analyzed to identify multiple research methods. Qualitative research methods presented a clear preference for the articles ($n=44$), consisting mostly of case studies ($n=30$) and interviews ($n=11$). This indicates the novelty of the research field through inductive exploratory research (Gioia et al. 2013). Approximately one-fifth of the articles ($n=12$) used quantitative methods including surveys and datasets.

Most of the articles from the sample have conducted their data collection in either European countries ($n=32$) or North America ($n=17$) (with the USA dominating with ($n=16$)). With other continents being underrepresented comparatively (Table 2). The

Table 2 Distribution of number and percentage of articles per continent

Continent	Number of articles	Percentage
Europe	33	51%
North America	17	26%
South America	5	8%
Asia	4	6%
Africa	1	1%
Multiple Countries	5	8%

reasons for this could be the use of keywords and selection criteria searching for articles specifically related to intermediaries in SFSC literature, written in scientific journals in English, excluding gray literature or region-specific case studies published in non-indexed outlets in languages other than English.

Results

While the literature describes intermediaries using different names, the authors grouped intermediaries together based on their common organizational structures. In this way, seven different intermediaries are presented and their capabilities investigated (Table 3). Within AFN literature, of which SFSC are a part of, researchers often talk about alternative and conventional intermediaries. Alternative intermediaries are considered mission-driven intermediaries that aim to create value in the entire SC and try to accomplish socioeconomic as well as socio-environmental sustainability in local communities (Jarzebowski et al. 2020; Alberti et al. 2024; Thomas et al. 2024). At the same time, conventional intermediaries often prioritize economic efficiency and productivity and are thus capable of delivering large volumes of food to meet customer demand (Thomas et al. 2024). The alternative intermediaries recognized in this section are producer cooperatives, consumer cooperatives, farmers markets and food hubs. The conventional intermediaries are retailers and wholesalers.

The results show a wide range of capabilities contributing to value creation in SFSCs. They range from more traditional wholesale functions related to operations management including aggregation and distribution, to more value adding functions such as producer support, including direct investments, as well as community engagement involving the society as a whole. An overview of the capabilities per intermediary that can be derived from the literature is found in Table 4. Where there is no cross, the capability was not

Table 3 Identified intermediaries from the sample

Intermediary	References
Producer cooperatives	(Badraoui et al. 2020; Ciliberti et al. 2020; Hu and Zhang 2024; Poças Ribeiro et al. 2021; Rosol and Barbosa 2021; Schreiber et al. 2023)
Consumer cooperatives	(Brunori et al. 2012; Vittersø et al. 2019; Carlson and Bitsch 2019; Kurtsal et al. 2020; Ajates 2021; Alberio and Moralli 2021; Poças Ribeiro et al. 2021; Rosol and Barbosa 2021; Ruszkai et al. 2021; Kump and Fikar 2021; Fuentes and Fuentes 2022; Hunter et al. 2022; Leipamaa-Leskinen et al. 2022; Benedek 2023)
Farmers markets	(Chiffoleau et al. 2016; Rover et al. 2017; Benedek et al. 2018; Loconto et al. 2018; Kurtsal et al. 2020; Alberio and Moralli 2021; Escobar-López et al. 2021; Rosol and Barbosa 2021; Vázquez and del Moral 2022; Pasquier Merino et al. 2022; Belletti et al. 2024; Monticone et al. 2024)
Retailers	(Ilbery et al. 2004; Carvalho de Rezende 2013; Syrovátková 2016; Malak-Rawlikowska et al. 2019; Vittersø et al. 2019; Kurtsal et al. 2020; Rosol and Barbosa 2021; Thomas et al. 2024)
Business platforms	(Loconto et al. 2018; Kurtsal et al. 2020; Rosol and Barbosa 2021; Ruszkai et al. 2021; Stephens and Barbier 2021; Hunter et al. 2022; Saucède and Vidal 2024)
Wholesalers	(Ilbery et al. 2004; Dimitri and Oberholtzer 2009; Bloom and Hinrichs 2011; Conner et al. 2014; Farmer and Betz 2016; Brekken et al. 2017; Grando et al. 2017; Roy et al. 2019; Valencia et al. 2019; Cardona et al. 2021)
Food hubs	(Dimitri and Oberholtzer 2009; Rogers and Fraszczak 2014; Cleveland et al. 2014; Clark and Inwood 2016; Engelseth 2016; Paciarotti and Torregiani 2018; Tewari et al. 2018; Furman and Papavasiliou 2018; Christensen et al. 2019; Trivette 2019; Winarno et al. 2020; Pesci and Brinkley 2021; Havadi-Nagy 2021; Anggraeni et al. 2022; Cunningham et al. 2022; Hyland and Macken-Walsh 2022; Kawasaki et al. 2023)

Table 4 Capabilities per intermediary

Category	Capabilities	PC	CC	FM	BP	FH	RT	WS
Operations management	Warehousing	X	–	–	X	X	X	X
	Transportation from producers	X	X	X	X	X	X	X
	Transportation to customers	X	X	X	X	X	X	X
	Develop distribution channels	X	X	X	X	X	X	X
	Aggregation	–	–	–	–	X	X	X
	(Re-)Packing	–	X	X	X	X	–	X
	Processing	–	–	X	X	X	–	–
	Quality and safety control	X	X	X	X	X	X	X
	Waste management	X	X	X	X	X	X	X
	Storage	X	–	–	–	X	X	X
	Order management	X	X	X	X	X	X	X
	Match supply and demand on local level	X	X	X	X	X	X	X
	Ensure consistent supply	X	X	X	X	X	X	X
	Negotiating with buyers	X	X	X	X	X	X	X
Marketing	Invoicing	X	X	X	X	X	X	X
	Price setting	X	X	X	X	X	X	X
	Develop sales methods	X	X	X	X	X	X	X
	Marketing communications	X	X	X	X	X	X	X
	Find new markets/customers	X	X	X	X	X	X	X
	Create demand for specialty products	X	X	X	X	X	X	X
Producer support	Collaborative production planning	X	X	–	X	X	–	–
	Investment in producer capacities	X	X	–	–	X	–	–
	Development of innovative products	X	X	–	–	–	–	–
	Recruitment of producers	X	X	X	X	X	X	X
	Facilitate communication between producers	X	X	X	X	X	–	–
	Coordinate production standards in SFSC	X	X	X	X	X	–	–
	Technical support for certification	X	X	X	–	X	–	–
Community engagement	Educational programming for customers	–	X	–	–	X	–	–
	Facilitate communication between stakeholders SFSC	X	X	X	X	X	X	X
	Innovation brokers SFSC	X	X	–	–	X	–	–
	Organize farm visits	X	X	–	–	X	–	–
	Institutional recombination (i.e., linking institutions and resources in new ways)	X	X	–	–	X	–	–
	Encourage training, innovation and new technologies	X	X	–	–	X	–	–
	Focus on environmental practices	X	X	X	X	X	X	X
	Ensure healthy food for low-income and urban communities	X	X	X	–	–	–	–
	Create new jobs	X	X	X	X	X	X	X

described in the literature sample in this review. A description of the different intermediaries and their capabilities is found in the next sections.

Producer cooperatives

The SFSC literature describes mostly direct-to-consumer producer initiatives, where producers differentiate their services by delivering directly to consumers through home deliveries and farmers' shops (Poças Ribeiro et al. 2021). The producers are solely

responsible for the processing and sales of their products, in accordance with direct-to-consumer SFSC logic. The limitations of producers to manage their usual operations while providing a wide range of services, in addition to reaching out to large customers and consumers, are leading to unsustainable conditions for the producers. To overcome these limitations, the literature recognizes situations in which producers and distributors have come together to form an intermediary organization through a collaborative association creating producer cooperatives (Schreiber et al. 2023).

Producer cooperatives operate as intermediaries in SFSCs when a group of producers and distributors sources products from multiple member-farms and organizes a collaborative form of intermediary, where two or more producers work together to serve a common set of customers (Badraoui et al. 2020; Rosol and Barbosa 2021; Schreiber et al. 2023). The organization and strategies are developed within the cooperative, acting as a unified intermediary competing with larger stakeholders in CFSC (Badraoui et al. 2020; Ciliberti et al. 2020). Producers' motivation to take part in such cooperatives depends on the specialization of crops (where some forms of agriculture are costlier than others), human capital and network (Ciliberti et al. 2020; Schreiber et al. 2023). While the cooperative exists as an independent organization, commitment and trust while maintaining property rights and autonomy, are important factors for small and medium-sized producers to maintain (Badraoui et al. 2020; Ciliberti et al. 2020).

The advantages of horizontal collaboration among producers in a SC have therefore been described as an innovative way for companies to improve their competitive performance in different ways.

From an RBV, producer cooperatives combine individual members' resources into collective capabilities that strengthen their position within SFSCs. Tangible resources, such as warehouses, transportation, equipment and processing facilities, are shared, lowering individual costs and investments. Access to financing improves as risks are spread across multiple farms. Human capital is leveraged through shared labor and knowledge exchange, enabling members to adopt best practices and respond more flexibly to operational demands (Badraoui et al. 2020; Hu and Zhang 2024).

While producer cooperatives increase competitive advantage through clear SC and marketing capabilities, their services are what makes them unique forms of SFSCs. Consisting of mainly producers, producer cooperatives support each other in numerous ways. Organizational capabilities emerge through collaborative production planning, coordinated logistics and joint marketing strategies, which allow the cooperative to meet larger or more consistent demand and enhance visibility in the marketplace. Relational capabilities are strengthened by expanding market access, improving collective bargaining power and representing producers' common interests before institutions and governments (Schreiber et al. 2023).

In addition, producer cooperatives develop social capabilities by fostering trust among members, engaging with local communities and organizing events that strengthen producer and consumer relationships. Taken together, these shared resources and capabilities provide a competitive advantage by increasing efficiency, reducing risks and improving resilience to disruptions (Schreiber et al. 2023). The cooperative has furthermore a clear focus on community engagement by organizing events that link the community to the cooperative in various ways (Hu and Zhang 2024).

With regard to sustainability, challenges of producer cooperatives are well described in Hu and Zhang (2024) where in many cases, the producer cooperatives have an important focus on economic sustainability through agricultural intensification and commercialization, while social and ecological sustainability are often sacrificed as trade-offs.

Consumer cooperatives

Consumer cooperatives can exist in different forms and act as intermediaries in SFSCs in different ways. The types of consumer cooperatives identified from the sample are Food community networks (FCNs) (Kurtsal et al. 2020; Ajates 2021), grassroot initiatives (Kump and Fikar 2021) and consumer-driven food corporations, such as REKO-ringen (Rosol and Barbosa 2021; Fuentes and Fuentes 2022). Overall, consumer cooperatives are driven by consumers in the form of civil society organizations (Kurtsal et al. 2020; Ajates 2021), or collaborative engagement platforms (Leipamaa-Leskinen et al. 2022). At their core, consumer cooperatives are constructed to enable the sale of local food products directly from producers to consumers, in a sustainable way (Kurtsal et al. 2020; Ajates 2021; Fuentes and Fuentes 2022; Benedek 2023). What they have in common is that they are considered third-party organizations when the cooperatives are formalized in some way and all participants obey by a fixed set of rules that each individual member should not deviate from.

While consumer cooperatives are often considered direct-to-consumer SFSCs, that should enable direct-to-consumer transactions, they are considered an intermediary consumer cooperation when the consumer market is relatively large, both in number and geographical distance. Here, the relationship between consumers and producers is enabled, shaped and maintained by consumer cooperatives (Alberio and Moralli 2021).

From a RBV, the unique capabilities of consumer cooperatives lie in how they transform member resources into collective forms of organization, market access and resilience. Their main tangible resources are limited, as most cooperatives do not own warehouses or equipment, but instead operate through digital platforms that coordinate sales, distribution, marketing and communication (Fuentes and Fuentes 2022; Hunter et al. 2022; Leipamaa-Leskinen et al. 2022). These platforms act as shared technological resources that reduce transaction costs and extend the reach of local food systems. Human resources are central, and members often contribute on a voluntary basis, taking turns in key tasks such as distribution, weighing products, communication with producers, finances and marketing (Ajates 2021; Fuentes and Fuentes 2022; Kump and Fikar 2021; Kurtsal et al. 2020; Rosol and Barbosa 2021). Through this pooling of labor and skills, consumer cooperatives create organizational capabilities for self-management and democratic governance.

Relational and social resources are particularly important. Cooperatives build trust between producers and consumers by guaranteeing product quality through dialogue, digital transparency (pictures, updates) and face-to-face interactions at drop-off points (Fuentes and Fuentes 2022; Leipamaa-Leskinen et al. 2022). These mechanisms strengthen relational capabilities by fostering commitment and accountability across the supply chain. Moreover, consumer cooperatives mobilize social capital by encouraging solidarity, collective identity and community engagement, thereby ensuring member loyalty and long-term participation.

In more formalized cooperatives, such as Solidarity Purchasing Groups (SPGs), these relational resources are further structured through regulated routines and shared responsibilities, enhancing coordination and continuity (Brunori et al. 2012; Carlson and Bitsch 2019; Vittersø et al. 2019). The strategic aims of SPGs include reflexive consumption, pursuing social justice, environmental sustainability and quality of specialized products. The aim is to achieve solidarity within the group and with producers aimed at improving employment and working conditions, as well as sharing ideas and using social links to improve economic sustainability in both food production and distribution (Brunori et al. 2012; Carlson and Bitsch 2019; Vittersø et al. 2019). Their functions and capabilities are the same as the non-formalized consumer cooperatives, with the difference that SPGs often have more regulated routines and responsibilities demanding a different level of involvement of its members then the more informal consumer cooperatives.

Third parties organizing farmers' markets

From the types of SFSCs, farmers markets (FMs) are probably the most known and are generally considered direct-to-consumer SFSCs, which in some cases had been regulated as well (Benedek et al. 2018; Pasquier Merino et al. 2022; Monticone et al. 2024). However, in the case that FMs are organized by a third party, their main function is mediation between producer and consumer, and through their governance structure this third party FM could be considered an intermediary (Belletti et al. 2024; Vázquez and del Moral 2022). Motivations behind FMs often give small producers an opportunity to sell their products directly to consumers and to create a marketplace for specific products (Kurtsal et al. 2020; Pasquier Merino et al. 2022; Rosol and Barbosa 2021). FMs are recognized as promoters of social innovation through the activation of community ties (Monticone et al. 2024). This form of social sustainability is gained through increased sales of fresh and healthy products in rural areas and to urban consumers, as well as the revival of rural areas through an increased focus on workers' rights and support of small-scale agriculture (Loconto et al. 2018; Belletti et al. 2024). Both material and immaterial (values and commitment) resources are important to both the producer and consumer sides to make this market work (Belletti et al. 2024; Vázquez and del Moral 2022). Additionally, collaboration and support are often necessary from governmental institutions, such as municipalities, who provide space for the market (Chiffoleau et al. 2016; Loconto et al. 2018; Kurtsal et al. 2020; Pasquier Merino et al. 2022).

Unique capabilities of FMs are their ability to conduct quality control of products, establish trust among producers and consumers, and provide training for producers (Chiffoleau et al. 2016; Escobar-López et al. 2021; Kurtsal et al. 2020). These actions are often managed by a third-party organization that can be formed by public-led institutions, producers or consumer organizations (Loconto et al. 2018).

The difference between FMs and the producer and consumer cooperations mentioned earlier is the importance of formal governance, and often the use of a trademark or certification to support quality and create trust (Alberio and Moralli 2021; Chiffoleau et al. 2016; Escobar-López et al. 2021). In some studies, these forms of certification are organized under the name participatory guarantee systems (PGSs), where the label added to the SFSC functions as a brand and quality guarantee (Alberio and Moralli 2021; Rover

et al. 2017; Vázquez and del Moral 2022). Furthermore, third-party organizations control, to a certain extent, the production, selling and adequate organization of markets, ensuring sufficient demand and supply (Alberio and Moralli 2021; Escobar-López et al. 2021; Rover et al. 2017).

Business platforms

Nowadays, some intermediaries in SFSCs function mainly online by creating a digital network or platform to connect local producers and buyers (Hunter et al. 2022; Kurtsal et al. 2020; Rosol and Barbosa 2021; Ruszkai et al. 2021). Producers using digital platforms operate simultaneously in a digital dimension, providing the customer with information at a distance, which leads to geographical and organizational proximity by the direct delivery of products and close interactions (Stephens and Barbier 2021).

Business platforms offer an alternative to the traditional brick-and-mortar retailers and exist in different forms and are known under different names (Saucède and Vidal 2024). What they have in common is that it is a platform controlled by a third party, which can be a producer or consumer group, volunteer, public-led organization or for-profit retailer. While business platforms resemble retailers in certain ways, they represent a distinct intermediation model. They typically do not take ownership of products but instead rely on digital infrastructures to facilitate interactions between SFSC stakeholders (Saucède and Vidal 2024).

From a RBV the value of these platforms lies in their ability to combine tangible and intangible resources into distinctive capabilities. Capabilities of business platforms can encompass traditional SC operations, enabling distribution, price setting, as well as hedonistic value creation for customers through increased direct communication between producers and consumers (Saucède and Vidal 2024; Stephens and Barbier 2021). These kind of initiatives aim to increase market diversity for different products (Loconto et al. 2018).

The technological resources of the platform such as digital infrastructure and online payment systems enable communication, coordination and transparency between producers and consumers (Hunter et al. 2022; Kurtsal et al. 2020; Loconto et al. 2018; Rosol and Barbosa 2021; Ruszkai et al. 2021). The pricing strategies of the platforms can be transparent and fixed, allowing for shared value creation in the SFSCs (Saucède and Vidal 2024; Stephens and Barbier 2021). Owners behind business platforms can fulfill a wide range of services, such as developing digital solutions, defining rules and guidelines, controlling membership and finding customers. Their ability to organize both operational, marketing and support functions are valuable intangible resources contributing to the competitive advantage of the platform (Hunter et al. 2022; Kurtsal et al. 2020; Loconto et al. 2018; Rosol and Barbosa 2021; Ruszkai et al. 2021).

Food hubs

One intermediary that rose to prominence in recent SFSC literature is the food hub, which is corroborated by appearing in 17 articles from the sample (total sample $n=65$). Food hubs can exist in various forms. The literature describes them mainly as a type of wholesaler, but depending on the geographical location of the study, they have also appeared as producers' distribution platforms (Rogers and Fraszczak 2014) or formalized

joint-stock company (Kawasaki et al. 2023). Food hubs mainly function as local hubs that connect different producers with different customers, such as retailers, schools, hospitals and restaurants, acting as selling agents (Cleveland et al. 2014; Engelseth 2016; Furman and Papavasiliou 2018; Trivette 2019), but also attract consumers by offering a diverse set of products fulfilling the consumer market's needs (Hyland and Macken-Walsh 2022).

Food hubs in SFSCs are often mission-driven intermediaries that aggregate products, distribute value-added foods and provide a wide range of value-added services such as coordinating production standards, organizing educational programming for consumers and investing in producer capacities (Clark and Inwood 2016; Tewari et al. 2018). They differ from conventional wholesalers in that their operations mix business objectives with mission-driven objectives related to community development, health and education (Tewari et al. 2018; Trivette 2019).

Food hubs are intermediaries that can play an important role in overcoming many of the barriers associated with direct-to-consumer sales, including consistent supply, large-scale food distribution, facilitating communication, food safety assurance and solving quality control (Anggraeni et al. 2022; Christensen et al. 2019; Havadi-Nagy 2021; Kawasaki et al. 2023).

The food hubs relational capabilities are apparent as service organizations involved in the horizontal and vertical integration of the SC. They work upstream and downstream with stakeholders in the SC and are, therefore, the first to detect and react to issues in SFSCs (Cunningham et al. 2022; Dimitri and Oberholtzer 2009; Hyland and Macken-Walsh 2022; Winarno et al. 2020). Issues in SFSCs include limited availability for producers, weak relationships between producers and customers, lack of communication and difficulties in logistics, transportation and delivery services (Clark and Inwood 2016; Paciarotti and Torregiani 2018). Food hubs can solve these issues by enabling increased cooperation and information sharing between stakeholders in SFSCs as well as arranging logistics and distribution services (Paciarotti and Torregiani 2018; Pesci and Brinkley 2021). This will work to ensure a stable supply and demand for local products, ensure traceability of local products, work actively toward reducing food loss and waste in the SFSCs, and in some cases process food (Kawasaki et al. 2023). Food hubs can provide key distribution support for mid-scale producers with businesses that are too large to exclusively utilize direct-to-consumer markets and too small to access conventional distribution channels (Christensen et al. 2019; Clark and Inwood 2016). As trust and commitment are key to SFSCs' survival, it is important that food hubs are proficient networking agents (Engelseth 2016; Pesci and Brinkley 2021).

Retailers

Some research in the sample showed that the most common intermediaries in SFSCs are retailers. Sales via conventional retailers make out the relative highest volume sold and market share in SFSCs, even if they do not necessarily include the highest number of producers (Malak-Rawlikowska et al. 2019). Retail shops in the literature include a wide variety of both alternative retailers such as local shops, farm shops, specialty shops, as well as conventional retailers such as supermarkets (Kurtsal et al. 2020; Rosol and Barbosa 2021; Syrovátková 2016; Vittersø et al. 2019). Of these retailers, farm shops in

which the producer sells their products directly to the consumer are generally considered as a direct-to-consumer SFSC. In the case of intermediate SFSCs, farm shops usually operate as middlemen by reselling goods produced by a number of producers and express significant awareness of the sustainability principles of the local food concept (Syrovátková 2016).

Capabilities creating competitive advantage lie mainly in the operational and marketing functions of retailers. They aggregate and sell products, keep stock and develop marketing and sales techniques. Their role in supporting producers and engaging the community are limited. Retailers can fulfill an intermediary role in communication by trying to create a relationship between consumers and producers. The main difference between the alternative and conventional retailers lies in the collaborative relationships between producers and retailers which appear to be stronger in the alternative retailers, such as local shops and farm shops, compared to conventional retailers (Alberti et al. 2024; Thomas et al. 2024).

Alternative retailers face challenges moving large volumes of producer through the SC due to distribution difficulties, as well as their instability to stock produce due to restricted space and warehousing capabilities. Here, conventional retailers can provide a solution to otherwise unstable market construction through their established distribution network and warehousing (Carvalho de Rezende 2013; Thomas et al. 2024). The power of conventional retailers has grown in recent years, as large retailers try to affect local food by adopting products from alternative chains and associating their own brands to this (Carvalho de Rezende 2013).

Wholesalers

With increased interest in locally produced food in the consumer market, conventional wholesalers respond to the opportunity to increase local food sales by using their existing SC infrastructure (Bloom and Hinrichs 2011; Roy et al. 2019). Their suppliers are often medium-sized producers that have enough produce to fulfill the market's demand but experience barriers in handling food safety costs, lack of market SC partners, adequate infrastructure, and lack of network and support (Brekken et al. 2017; Dimitri and Oberholtzer 2009; Farmer and Betz 2016).

Wholesalers in the context of SFSC are almost often conventional intermediaries that manage a demand-driven SC, where customer demand and financial profits come first over relationships with local producers (Bloom and Hinrichs 2011; Roy et al. 2019). However, in some SFSCs, rural wholesalers are presented having intergenerational social ties with producers where trust has been created over the years. The strength of these relationships serves as an incentive for wholesalers to remain honest, reliable and responsive to potential problems (Bloom and Hinrichs 2011; Roy et al. 2019).

In intermediate SFSCs, wholesalers serve as important mediators between producers and large customers, such as institutional procurement (Conner et al. 2014). In this type of SFSCs when public procurement is involved, all stakeholders tend to work toward shared profits without negatively affecting the social values of supporting local producers (Conner et al. 2014). When this becomes difficult, a solution could be to introduce public support in the form of public procurement, which could provide stability in the SC (Valencia et al. 2019).

Overall, wholesalers fulfill several functions in SFSCs. They connect supply and demand (Brekken et al. 2017; Roy et al. 2019). They assist local producers in packaging and labeling products (Grando et al. 2017). They take care of their distribution (Cardona et al. 2021; Grando et al. 2017). They facilitate cooperation between stakeholders in the entire SC (Roy et al. 2019; Cardona et al. 2021) and can create and develop new ways of sales by, for example, box schemes, where multiple products from different producers are collected and distributed to a customer base (Cardona et al. 2021).

Discussion

In this review, the authors extracted and analyzed information from 65 articles related to intermediate SFSC in order to answer the research questions. The articles were analyzed in two steps, first the authors identified and categorized the different intermediaries operating in SFSCs, then the authors mapped these companies' capabilities to create value in SFSCs and obtain competitive advantage toward CFSCs.

Intermediaries in SFSCs

Even if the literature discusses SFSC as the most sustainable if the number of intermediaries is reduced, the shortfall of key intermediaries (e.g., abattoirs, transporters, wholesalers) threatens food production in rural areas, as producers still need to connect with key "external" stakeholders in order to serve the market (Ilbery et al. 2004). Intermediaries provide solutions for medium-sized producers, who may be too large or otherwise unsuitable for direct markets and offer opportunities for smaller producers to aggregate their products to reach larger markets (Peterson et al. 2021). However, for intermediaries to create value for all stakeholders in SFSCs, their capabilities should match the strong values that are so prominent in SFSCs.

Between direct-to-consumer and intermediate SFSC, it is found that depending on the number of producers, number of intermediaries, unit transportation cost and producers' culture, intermediaries operating in intermediate SFSCs cause greater traded quantities than direct-to-consumer sales (Mejia and Garcia-Diaz 2018). Previous research has shown that intermediation helps shorten negotiation times, matches customers with producers and reduces transaction costs (Mejia and Garcia-Diaz 2018). Intermediaries in SFSC have a special role, where they not always or exclusively operate as market intermediaries, but rather as providers of a range of services within food systems. They are important stakeholders whose activities affect the nature and success of market exchanges (Loconto et al. 2018). Even if intermediaries generally function as external third parties and for-profit organizations, intermediaries in SFSC can also be formalized as nonprofit organizations, which include associations or consumer cooperatives and foundations (Poças Ribeiro et al. 2021). Some research has found that within some intermediate SFSC cooperation models, different stakeholders work together that do not necessarily originate from the agriculture sector (Martens et al. 2022).

The review recognizes alternative and conventional intermediaries operating in SFSCs. One of the main differences between these intermediaries lie in their level of formalization. In highly relational and social initiatives such as producer and consumer cooperatives, the literature recognizes a low level of formalization. These types of intermediaries differ from direct-to-consumer SFSC in that direct transactions between producers and

consumers are enabled by different platforms, such as digital platforms, managed by a few volunteers, or retailers, including local shops, and provide their services without compensation (Hunter et al. 2022; Kump and Fikar 2021; Kurtsal et al. 2020).

In intermediate SFSCs that include intermediaries such as producer cooperatives, farmers' markets, retailers, wholesalers and food hubs, this review identifies more formalized collaboration via contracts and communication, including the use of a trademark or certification. This is important for supporting quality and creating trust between the intermediary and the remaining stakeholders in the SC (Alberio and Moralli 2021; Chiffoleau et al. 2016; Escobar-López et al. 2021).

The sample shows that many intermediate SFSCs are dynamic and adopt a continuously changing environment. This became clear by looking at consumer cooperatives, which can start off small-scale and local, but might diffuse into larger regional food systems over time by growing in scale and replicating at new locations or even change into a different intermediary where other intermediaries such as retailers adopt some features from these consumer cooperatives (Kump and Fikar 2021). The need to be dynamic and scale up operations to become or remain competitive advantage, comes with challenges. Social initiatives, such as consumer and producer cooperatives, can feel pressurized to focus on growth and profitability through agricultural intensification and commercialization, adapting more CFSC behaviors, sacrificing the social, relational and ecological values along the way (Hu and Zhang 2024).

Capabilities creating value in SFSCs

The main difference between intermediaries operating in CFSC and SFSCs is that intermediaries in CFSC are often solely motivated by the business opportunity of intermediation, whereas intermediaries involved in SFSCs mix both business objectives and mission-driven objectives in their operations (Mejia and Garcia-Diaz 2018; Tewari et al. 2018). Relationships and trust among all stakeholders are an important part of the SFSC (Hunter et al. 2022; Rosol and Barbosa 2021).

Most of the intermediaries' capabilities presented in this review fulfill basic SC functions, such as aggregation of products, sales and distribution. However, the alternative intermediaries (i.e., producer- and consumer cooperatives, farmers markets, business platforms and food hubs) show more relational capabilities related to producer support and community engagement, compared to the conventional intermediaries (i.e., wholesalers and retailers) (Table 4).

The focus on relationships and social resources is the strongest in producer- and consumer cooperatives, as well as farmers markets (Fig. 4). Besides the common operational functions related to operations management and marketing, these cooperatives are unique in the sense that from all the intermediaries they have the largest focus on producer support and community engagement by working on a small and local scale, increasing the sense of community in the SC. This has materialized through the organization of social events such as workshops, seminars and educational events (Kump and Fikar 2021; Kurtsal et al. 2020). This review highlights, using a RBV perspective, that within consumer cooperatives the unique capabilities lie in intangible, relational and

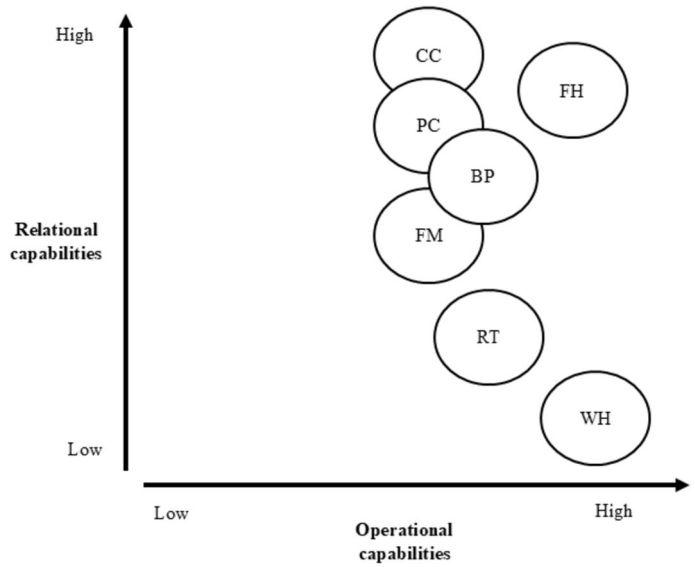


Fig. 4 Relational and operational capabilities per intermediary (own elaboration)

human resources, whereas producer cooperatives rely more heavily on tangible and production-related resources.

To obtain a more sustainable food system, the number of- and sales through SFSCs should increase. The big question lies in which intermediary is capable of accomplishing this in a sustainable way. Currently, the conventional intermediaries (wholesalers and retailers) make up the largest sales and volumes in SFSCs. So, if the answer lies with them, these intermediaries must be able to go beyond simply satisfying customer demands, considering the vital role of social values in SFSCs. They are furthermore expected to uphold the fundamental principles of SFSCs, with a strong emphasis on promoting producer engagement and empowerment. As relationships between stakeholders in the SC are important, the capabilities of the conventional intermediary may have to be directed in a way that ensures a sustainable outcome for the entire SC (Kump and Fikar 2021). While this is an ideological stance, in reality most conventional intermediaries manage a demand-driven SC, where consumer demand and financial profits come first over relationships with local producers (Bloom and Hinrichs 2011).

Directing our attention to alternative intermediaries, one can recognize that while presenting a romanticized view of sustainable SCs, some of these intermediaries like producer- and consumer cooperatives are informal in nature and are struggling to compete with the established conventional intermediaries. Even if these intermediaries are represented in the literature sample as an alternative solution to conventional intermediaries, one could argue that in their current form they mainly focus on proximate SFSC rather than spatially extended SFSCs. This implies they are serving a very local market, with limited sales and customers, therewith not creating a strong competitive advantage against the conventional intermediaries. Business platforms, on the other hand, provide a solution to the locality of these initiatives by exploiting digital platforms to reach a larger audience. They provide solutions of practical nature

including collaborative logistics and distribution, as well as support with sales and invoicing (Saucède and Vidal 2024). However, when these initiatives grow, one recognizes an increased level of formalization that impacts governance, transparency and equity in the initiatives. In some literature, it appeared that some of these initiatives ultimately developed into a different kind of intermediary such as a local shop, i.e., a retailer, leading to some disconnection between stakeholders in the SC (Kump and Fikar 2021).

Strategically, these pooled resources generate capabilities that differentiate consumer cooperatives from conventional intermediaries: democratic governance ensures collective decision-making; digital platforms provide operational efficiency and scalability; social ties create resilience against disruptions; and solidarity principles align consumer demand with broader goals of social justice, sustainability and fair working conditions. By turning individual contributions into shared organizational strength, consumer cooperatives create a competitive advantage based not on capital-intensive infrastructure, but on community, collaboration and values-driven consumption.

One alternative intermediary that appeared as a counterbalance to conventional wholesalers and retailers, and which might serve relatively larger markets in both proximate and spatially extended SFSCs are food hubs. Food hubs are closely related to wholesalers and retailers and operate as key intermediaries in operations management capabilities, such as aggregation and distribution of products, but in addition focus on mission-driven activities, such as the education of consumers and investing in producer capacities. Furthermore, they serve both small private customers, such as consumers, and larger business customers, such as retailers, restaurants and public procurement organizations such as schools and hospitals. Their sales efforts and customer focus might potentially lead to larger volumes of products being sold through the SFSCs.

Conclusion

For SFSCs to serve as a sustainable alternative to CFSCs, they must increase both in number and in the volume of products moved through the SC. Understanding how SFSCs can achieve growth and secure a competitive advantage relative to CFSCs requires an examination of the intermediaries that play a central role in these chains. To this end, the authors analyzed seven intermediaries and their capabilities through the lens of a RBV framework.

While the authors acknowledge that there might be more types of intermediaries existing in many forms that are difficult to define, the categorization presents a broad overview allowing for a more thorough investigation of the SC capabilities creating value in SFSCs.

The findings show that conventional intermediaries such as retailers and wholesalers are the most common intermediaries in SFSC, with the highest volume sold and market share (Malak-Rawlikowska et al. 2019). Their business models are, however, mostly demand-driven and do not necessarily align with the social values present in SFSCs. It is therefore questionable if they are in their current form, able to lead the way in transforming the food system from CFSC to SFSCs.

The results illustrate a difference between conventional and alternative intermediaries and their capability of creating value in SFSCs. While all intermediaries are involved in

operations management, the alternative intermediaries, such as producer cooperatives, consumer cooperatives, FMs, business platforms and food hubs, also have the capability of contributing to socioeconomic sustainability in SFSCS through providing producer support and community engagement.

The discussion around which intermediary is most suitable to increase the competitive advantage of SFSCs toward CFSC is a complicated one. While alternative intermediaries have capabilities aligning with the SFSCs values, these alternative intermediaries generally operate in very local markets, in proximate SFSCs, where small volumes of products are moved. For example, FMs are designed to give small producers a chance to sell their products directly to consumers, and to create a marketplace for specific products (Kurtsal et al. 2020; Rosol and Barbosa 2021). In addition, consumer cooperatives and business platforms target local communities as their primary customers. These are interesting to study, but these types of intermediaries might not lead to increased volumes to make any kind of impact on changing conventional food systems to more sustainable solutions. Looking at intermediaries that operate both in proximate SFSCs and spatially extended SFSCs could be one way of increasing volumes while maintaining the sustainability focus that is key to an SFSC.

Implications

The theoretical implications of this review are an overview over the seven most mentioned intermediaries in SFSC and their capabilities in creating value in SFSCs. It gives an overview of the common capabilities of intermediaries in SFSC that, at first, look different due to different contexts. It furthermore discusses the differences between these intermediaries and their capabilities in SFSC. The discussion focused in particular on the difference and importance of fulfilling not only traditional SC functions but also services that include producer support and community engagement, which can lead to a change to more sustainable food SCs.

The practical implications of this review are many and include a clear overview of the capabilities of different intermediaries that could contribute to competitive advantage of SFSCs. More broadly the outcomes of this review can inform policymakers and SFSC stakeholders in establishing appropriate intermediaries for the establishment and scaling of SFSCs. Examples could be that practitioners could choose to focus on the development of producer cooperatives because it might lead to competitive advantage through the sharing of resources, risk and access to financing. The development of consumer cooperatives as well as business platforms, could help rural areas extent their reach and could be accomplished by providing training for members and engaging the local community. The review showed that both material and immaterial resources are important for FMs to succeed, and governmental agencies can become aware of their role in enabling place and executing for these markets. Food hubs provided a possible alternative to conventional intermediaries and local funding schemes or innovation incentives for mission-driven intermediaries might encourage the development of these types of intermediaries. And finally, conventional stakeholders, such as wholesalers and retailers, could increase their focus on establishing more collaborative relationships with SFSC stakeholders to increase the sales through SFSC in a sustainable way.

Due to the apparent role of public procurement in the sustainability of SFSCs, these results could lead to the integration of intermediaries into local food policy and public procurement frameworks.

Limitations and further research

One of the limitations of this review lies in the use of scientific outlets only (i.e., WoS and Scopus), which has excluded gray literature, policy guidelines on local, national and international levels discussing SFSCs, as well as region-specific case studies published in non-indexed outlets.

In accordance with the SLR protocol, the findings reported in this article are derived from the outcomes of the specified methodology. Although the sample encompasses a broad range of studies, the inconsistent use of keywords within this expansive research field may have resulted in some relevant literature not being retrieved from the search databases. Future research could further investigate the diverse types of intermediaries and explore additional dimensions of these initiatives.

Future research directions could include a deeper look into every type of intermediary active in SFSC and delve into their business models to understand the capabilities in their context. Researchers could further investigate the drivers and barriers in establishing and operating this type of SFSCs, which could provide valuable knowledge to both theory and practice. This might also encourage other researchers to look closer to the role and functions of conventional intermediaries, such as retailers, in SFSCs.

The deductive coding structure informed by the RBV may have influenced the interpretation of the intermediary capabilities. The RBV only touches upon the relational aspects from a resource perspective, and additional theories such as network theories or stakeholder theories could provide a deeper understanding of the dynamic interactions and contextual factors influencing SFSCs.

Food hubs are presented as an interesting intermediary with the possibility of scaling up volumes through SFSCs, which originated as a response to the lack of mission-driven focus of conventional wholesalers. However, research with regard to this intermediary is mostly conducted in the USA. The suitability of this intermediary in other countries is therewith underexplored.

Furthermore, while business platforms are discussed in this review, the transformative potential of digital business platforms is underexplored in the sample. Digital tools are rapidly reshaping intermediation through numerous ways such as platform cooperativism, decentralized logistics and blockchain-enabled traceability. Further research could explore the use of digital platforms across different intermediaries to more thoroughly discuss its implications on sustainability and competitive advantage.

Finally, given the geographical limitations of most research in the sample being conducted in North America and Europe, future research could focus on the types of intermediaries and their capabilities in other regions.

Abbreviations

AFN	Alternative food networks
CFSC	Conventional food supply chain
RBV	Resource-based view

SC Supply chain
SFSC Short food supply chain

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